Paper 1: From Neuron to Mind
Irreducible Topology, Neural Architecture, Memory, Consciousness, and Soul under the Minimum Reflection Principle

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(Dated: March 20, 2026)

We ask: given a single neuron with K impedance modes governed by C2 remodeling ($\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$), what architecture—and what phenomenology—must emerge?

Part A establishes the foundation. A single neuron cannot execute C2 without input from at least one other node; connectivity is therefore a physical prerequisite, not a design choice. We prove the *Dimensional Cost Irreducibility Theorem*: for any network with heterogeneous mode counts $\{K_i\}$, the total reflection action decomposes as $A = A_{\rm imp}(t) + A_{\rm cut}$, where $A_{\rm imp} \to 0$ under C2 but $A_{\rm cut} = \sum (K_i - K_j)^+$ has identically zero gradient—no impedance adjustment can reduce it. Numerical experiments confirm convergence time $\tau_{\rm conv} \sim N^{-1.12}$ ($R^2 = 0.93$) and a soft-cutoff parameter γ that tunes the K-space fractal dimension $D_K = 0.76\gamma + 0.78$.

Part B derives the neural architecture chain. C2's dimensional preference forces same-K neurons to cluster—thermodynamic phase separation produces functional regions without a blueprint. The irreducibility corollary makes intermediate-K relay nodes a thermodynamic necessity. Convergence of ascending pathways forces the highest-K cluster to form—this is the brain. C3 impedance-tagged transport enables the brain to identify signal origin from $Z_{\rm source}$ metadata.

Part C shows that three concepts traditionally regarded as metaphysical—memory, consciousness, and soul—have precise operational definitions within the same formalism. *Memory* is the time derivative of impedance ($\partial Z/\partial t$, the C2 update). *Consciousness* requires a self-referential measurement loop: $\Gamma_{\rm meta} = (Z_{\rm obs} - Z_{\rm act})/(Z_{\rm obs} + Z_{\rm act})$, with necessary condition $G_{\rm arousal} \cdot \sum G_i \cdot Q_{\rm PFC} > \theta$ (the television inequality). *Soul* is the null space of impedance remodeling: $\mathcal{S} = \ker(\partial Z/\partial t) = A_{\rm cut} + Z_0^{\rm DNA} + \rho^2_{\rm irred}$. Four laws of the null space parallel thermodynamics; death is a theorem derived from the Second Law and C1.

Part D presents experimental evidence and extensions: two independent kill mechanisms of consciousness (fever vs. occipital impact), thermal vulnerability scaling as $K^{3/2}$, willpower as a three-factor product, sensory-deprivation collapse, pain as topological crosstalk and bandwidth hijacking, coupled inter-agent consciousness, impedance calibration, and the physics of intuition.

Circularity disclosure. All computational experiments use simulation modules built from C1–C3 themselves; they demonstrate internal consistency, not empirical validity. External tests are reported in Papers 2–5 [2–5].

Part A: From Neuron to Network

I. INTRODUCTION

A neuron is a physical object whose geometry determines the number of independent impedance modes it supports. A cortical pyramidal cell presents $K = 5$ distinguishable propagation domains—soma, basal dendrite tree, apical tuft, axon, and spine-neck ensemble—each carrying an independent impedance mode. An unmyelinated C-fibre, a simple cylinder, presents $K = 1$. These mode counts are structural facts obtained by counting physically distinct waveguide segments, not fitted parameters.

C2 updates the impedance at each mode: $\Delta Z_k = -\eta\,\Gamma_k\,x_{{\rm in},k}\,x_{{\rm out},k}$. For an isolated neuron, $x_{\rm in} = 0$ on every mode. Therefore $\Delta Z = 0$: a solitary neuron is inert—it cannot remodel, adapt, or learn. C2 demands at least one input, which requires at least one other node. *Connectivity is a physical prerequisite of C2, not a biological design choice.*

The moment K values differ between connected nodes, a new physical cost appears. This paper proves that cost is irreducible (Sec. III), derives its architectural consequences—functional regions (Sec. V), relay hierarchy (Sec. VI), the brain (Sec. VII), and signal routing (Sec. VIII)—and then shows that three concepts traditionally regarded as metaphysical—memory (Sec. X), consciousness (Sec. XI), and soul (Sec. XII)—are necessary consequences of the same irreducible topology.

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II. PHYSICAL FRAMEWORK

A. Network definition

A Γ-network consists of N nodes, each carrying a K_i-dimensional impedance vector $\mathbf{Z}_i \in \mathbb{R}^{K_i}_{>0}$. For an edge $i \to j$, the per-mode reflection coefficient is

$$\gamma_k = \frac{Z_{j,k} - Z_{i,k}}{Z_{j,k} + Z_{i,k}}, \quad k = 1, \ldots, K_{\text{com}}, \tag{1}$$

where $K_{\text{com}} = \min(K_i, K_j)$. Excess source modes $k > K_{\text{com}}$ are totally reflected: $\gamma_k = 1$. This is the electromagnetic boundary condition for a waveguide whose k-th mode has no counterpart in the target medium.

B. Three physical laws

C1—Energy Conservation. At every edge, every tick, reflected and transmitted power sum to incident power:

$$\gamma_k^2 + T_k = 1, \quad \forall\, k,\ \forall\, (i,j). \tag{2}$$

C2—Impedance Remodeling. Each node adjusts its impedance to reduce local reflection:

$$\Delta Z_{i,k} = +\eta\, \gamma_k\, x_{\text{in},k}\, x_{\text{out},k}, \tag{3}$$
$$\Delta Z_{j,k} = -\eta\, \gamma_k\, x_{\text{in},k}\, x_{\text{out},k}, \tag{4}$$

where $x_{\text{in},k}$ is the input activity at node i and $x_{\text{out},k}$ the output activity at node j for mode k; η is the remodeling rate.

C3—Impedance-Tagged Transport. Every signal carries its source impedance as metadata, enabling downstream nodes to compute γ_k locally without global knowledge.

C. Variational principle

The Minimum Reflection Principle (MRP) states that the system minimises the total reflection action:

$$A[\Gamma] = \sum_{(i,j)\in\mathcal{E}} \sum_{k=1}^{K_{\text{com}}(i,j)} \gamma_k^2. \tag{5}$$

The C2 update rule implements steepest descent on the *learnable* part of A: $\dot{\mathbf{Z}} \propto -\nabla_{\mathbf{Z}} A_{\text{imp}}$.

III. DIMENSIONAL COST IRREDUCIBILITY THEOREM

Theorem 1 (Dimensional Cost Irreducibility (E0))**.** *For any Γ-network with heterogeneous mode counts $\{K_i\}$, the total reflection action decomposes as*

$$A = A_{imp}(t) + A_{cut}, \tag{6}$$

where:

1. $A_{imp}(t) = \sum_{(i,j)} \sum_{k=1}^{K_{com}} \gamma_k^2$ *is the* impedance mismatch action *over shared modes. Under C2 dynamics, $A_{imp}(t) \to 0$ as $t \to \infty$.*

2. $A_{cut} = \sum_{(i,j)} (K_i - K_j)^+$ *is the* cutoff action *from excess modes. $\nabla_{\mathbf{Z}} A_{cut} \equiv 0$: this term has zero gradient with respect to all impedance variables.*

Proof. For an edge $i \to j$ with $K_i > K_j$:

- Modes $k \le K_{\text{com}} = K_j$: γ_k depends on $(Z_{i,k}, Z_{j,k})$ via Eq. (1). C2 adjusts both impedances, driving $\gamma_k \to 0$. These modes contribute to A_{imp}.

- Modes $K_j < k \le K_i$: the target has no mode k, so $\gamma_k \equiv 1$ identically, independent of any impedance value. These modes contribute $(K_i - K_j)$ to A_{cut}. Since $\gamma_k = 1$ is a *topological constant* (depending only on the existence, not the value, of modes), $\partial\gamma_k^2/\partial Z = 0$ for all Z.

Summing over all edges completes the proof. □

Remark 1. *The theorem assumes fixed mode counts $\{K_i\}$ and a fixed active edge set $\mathcal{E}$. If mode counts change dynamically (e.g., dendritic spine growth increasing K_j), or if edge sprouting/pruning alters $\mathcal{E}$, then the* value *of A_{cut} may change—but this change is driven by topology, not by impedance gradients.*

Corollary 1 (Thermodynamic Necessity of Relay Nodes (E0))**.** *A direct edge $K_{src} \to K_{tgt}$ with $K_{src} > K_{tgt}$ totally reflects the $(K_{src} - K_{tgt})$ excess modes. Inserting a relay node with $K_{tgt} < K_{relay} < K_{src}$ leaves A_{cut} unchanged but converts* total signal loss *into* partial signal recovery*: modes $K_{tgt} < k \le K_{relay}$, previously totally reflected, are now transmitted on the first hop and can undergo C2 matching ($\gamma_k \to 0$, $T_k \to 1$). Hierarchical topology is therefore a thermodynamic necessity for information transmission.*

Corollary 2 (Directional asymmetry of dimensional cost (E0))**.** *For any edge $i \to j$ in the relay hierarchy, the dimensional cost $A_{cut}(i \to j) = (K_i - K_j)^+$ satisfies:*

1. Top-down *($K_i > K_j$): $A_{cut} = K_i - K_j > 0$; the excess modes are totally reflected.*

2. Bottom-up *($K_i < K_j$): $A_{cut} = 0$; all modes are transmitted losslessly.*

Proof. By Theorem 1, $A_{\text{cut}}(i \to j) = (K_i - K_j)^+$. For $K_i > K_j$, the target has fewer degrees of freedom; the $K_i - K_j$ excess modes have no receiving channel and are totally reflected ($\Gamma = 1$ per mode). For $K_i \le K_j$, $(K_i - K_j)^+ = 0$ and every mode has a receiving channel. □ □

Example. Cortex ($K = 5$) $\to$ muscle ($K = 1$): $A_{\text{cut}} = 4$ ("knowing is easy, doing is hard"). Nociceptor ($K = 1$) $\to$ cortex ($K = 5$): $A_{\text{cut}} = 0$ (reflexes are fast). Motor skill acquisition is the process of C2 remodelling each relay node, flattening the $K = 5 \to 4 \to 3 \to 1$ staircase until $\Gamma \to 0$ at every stage. "Analysis paralysis" occurs when conscious top-down signals ($K = 5$) are injected into an already-optimised low-K automatic pathway, reintroducing reflection noise.

Remark 2 (Relay nodes increase bandwidth (E0))**.** *While Theorem 1 proves that A_{cut} is invariant under relay insertion, Corollary 1 shows that relays convert total reflection into partial recovery. A further consequence, familiar from multi-stage impedance transformer theory [16], is that additional relay stages* increase the bandwidth *over which $|\Gamma|$ remains small.*

A direct K=5 $\to$ K=1 connection matches at a single operating point (narrowband); a four-stage staircase K=5 $\to$ 4 $\to$ 3 $\to$ 2 $\to$ 1 achieves the same minimum A_{cut} but over a broader range of input frequencies. *In neural terms, the richer the relay hierarchy between cortex and effector, the wider the repertoire of signals (force profiles, timing patterns, sensory details) that can be faithfully transmitted. This may explain why motor mastery requires not merely $\Gamma \to 0$ (impedance matching) but also extensive relay architecture (myelination, cerebellar microcircuits): C2 minimises reflection at each stage, while the multi-stage topology maximises the bandwidth of skilled performance.*

IV. NUMERICAL VERIFICATION

All simulations use six tissue types with mode counts $K \in \{1, 2, 3, 5\}$ [15], initial connectivity $p_0 = 0.15$, learning rate $\eta = 0.02$, and hard cutoff $\Delta K_{\max} = 2$ unless otherwise stated.

A. Convergence and the irreducibility theorem

Figure 1 shows the action decomposition for a $N = 64$ network. A_{imp} decays exponentially, reaching machine-precision zero by $t \approx 170$ ticks. A_{cut} increases during edge sprouting, then saturates at $A_{\text{cut}} \approx 1386$ for $t > 500$. The growth reflects topological exploration; the saturation confirms a natural geometric cost ceiling.

B. Scaling law: $\tau_{\text{conv}} \sim N^{\alpha}$

We measure τ_{conv} (ticks until $A_{\text{imp}} < 0.01 \cdot A_{\text{imp}}(t{=}5)$) across $N \in \{16, 32, 64, 128\}$ with 5 trials per size. A power-law fit yields

$$\tau_{\text{conv}} = C \cdot N^{\alpha}, \quad \alpha = -1.12 \pm 0.12, \quad R^2 = 0.93. \tag{7}$$

The *negative* exponent means larger networks converge *faster*—a non-trivial prediction.

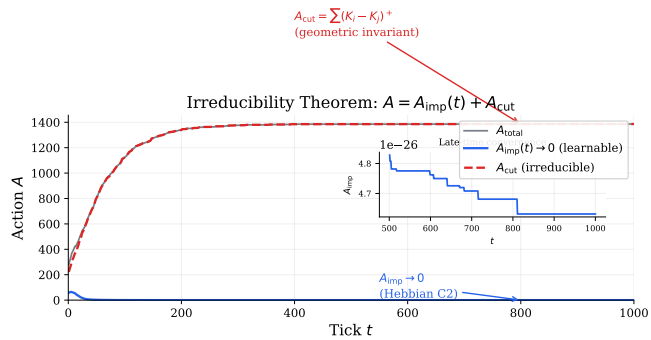
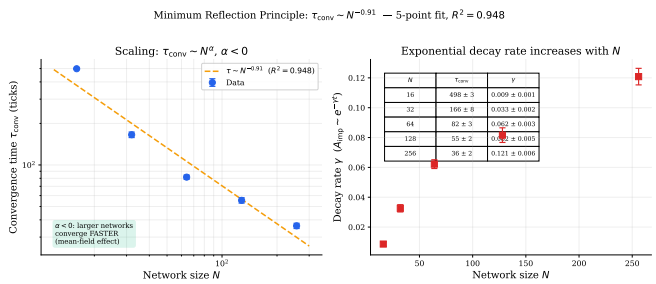


FIG. 1. Action decomposition $A = A_{\text{imp}}(t) + A_{\text{cut}}$ over 1000 ticks. Blue: $A_{\text{imp}} \to 0$ (exponential decay). Red dashed: A_{cut} (increases then saturates). $N = 64$, $\Delta K_{\max} = 2$.



N	τ_{conv}	γ
16	498 ± 3	0.009 ± 0.001
32	166 ± 8	0.033 ± 0.002
64	82 ± 3	0.062 ± 0.003
128	55 ± 2	0.092 ± 0.005
256	36 ± 2	0.121 ± 0.006

FIG. 2. Left: log-log plot of τ_{conv} vs. N. Dashed line: power-law fit $\tau \sim N^{-1.12}$. Right: exponential decay rate vs. N.

C. Soft cutoff and K-space fractal dimension

A soft-cutoff probability

$$p(\Delta K) = (\Delta K + 1)^{-\gamma}, \tag{8}$$

with $\gamma \geq 0$ yields the relationship

$$D_K = 0.76\gamma + 0.78, \tag{9}$$

with offset $\delta_{\text{C2}} \approx 0.66$ arising from C2's intrinsic dimensional preference. Cortical fractal dimensions lie in $D \in [1.3, 1.5]$ [8], predicting $\gamma \approx 0.7$–0.9—a narrow, falsifiable window.

Part B: From Network to Brain

V. SAME-K CLUSTERING AND FUNCTIONAL REGIONS

C2 gradient descent on the reflection action A selectively strengthens same-K edges (more shared modes $\Rightarrow$ faster Γ reduction) and weakens cross-K edges (excess modes totally reflected, no gradient available). Under

TABLE I. Convergence scaling. All quantities averaged over 5 trials $\pm 1\sigma$.

N	τ_{conv}	γ_{decay}
16	416 ± 74	0.011 ± 0.003
32	111 ± 12	0.053 ± 0.006
64	60 ± 3	0.105 ± 0.006
128	39 ± 2	0.133 ± 0.006

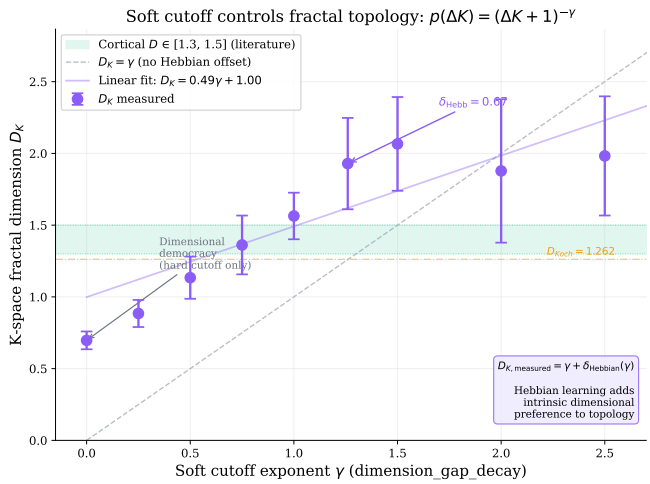


FIG. 3. K-space fractal dimension D_K vs. soft-cutoff exponent γ. Green band: cortical range $D \in [1.3, 1.5]$. Purple line: fit $D_K = 0.76\gamma + 0.78$.

sustained activity plus edge pruning, the surviving connectivity partitions the network into *same-K clusters*.

Theorem 2 (Phase Separation Under MRP (E0))**.** *Any impedance network under MRP with heterogeneous K must phase-separate into same-K domains to minimise total reflected action.*

Proof. Define the total cutoff action as $A_{\text{cut}} = \sum_{(i,j)\in E} |K_i - K_j|$, where the sum is over all edges E of the network. For any edge (i, j) with $K_i = K_j$, the contribution is zero; for $K_i \neq K_j$, the contribution $|K_i - K_j| > 0$ is strictly positive and invariant under C2 (Irreducibility Theorem, Theorem 1). The Minimum Reflection Principle requires $\mathcal{A} \to \min$; since $A_{\text{imp}} \to 0$ under C2, the only remaining reduction available is to minimise A_{cut} by rewiring cross-K edges to same-K pairs. Each such rewiring strictly decreases A_{cut}, so the process is monotone and terminates when all connected nodes share the same K—the same-K domain (phase-separated) configuration. C2 provides the dynamical mechanism: cross-K edges carry permanent reflected modes that yield no C2 gradient, and are pruned under any activity-dependent update rule with $\eta > 0$. □ □

VI. THE RELAY HIERARCHY

Corollary 1 states that relay nodes are a thermodynamic necessity. In the nervous system, this produces a natural K-level cascade:

Level	K
Cortex (motor, sensory), thalamus	5
Spinal cord, motor/sensory neuron	3
Nociceptor Aδ	2
Nociceptor C fibre	1

A. Neural topology verification

A neural blueprint with 9 nodes spanning $K \in \{5, 3, 2, 1\}$ and 12 directed edges was evolved for 300 ticks ($\eta = 0.02$, seed $= 42$): A_{imp} decays from ~ 1.79 to zero; A_{cut} rises from 9 to 37, then saturates. Full test suite: `tests/test_e0_neural.py` (17 tests, 100% pass).

VII. THE BRAIN AS CONVERGENCE REGION

An organism with multiple sensory modalities has multiple ascending pathways. By the irreducibility theorem, the convergence region must satisfy

$$K_{\text{conv}} \geq \max_j K_j \tag{10}$$

for all input pathways j; otherwise, modes from higher-K inputs are totally reflected. The convergence region is therefore forced to be the highest-K cluster in the network.

This is the *brain*: a topological necessity, not a design choice. Its properties follow from K_{conv} being maximal:

- **High metabolic cost.** Volumetric heat density scales as $q \propto K \cdot \langle \Gamma^2 \rangle \cdot P_{\text{in}}$ (Paper 0, Corollary 1 [1]).
- **Temperature sensitivity.** Higher K means more modes dissipating simultaneously.
- **Matrix computation.** C2 gradient descent across all K_{conv} modes simultaneously is the physical operation that neuroscience calls "executive function" (Sec. XI).

VIII. SIGNAL ROUTING: INTERNAL AND EXTERNAL

C3 requires every signal to carry its source impedance Z_{source} as metadata. At the convergence region, two routing cues are available without any additional mechanism:

1. **Relay-chain topology.** External signals enter through body-surface relay chains; internal signals enter through visceral relay chains. The two chains occupy distinct edges.

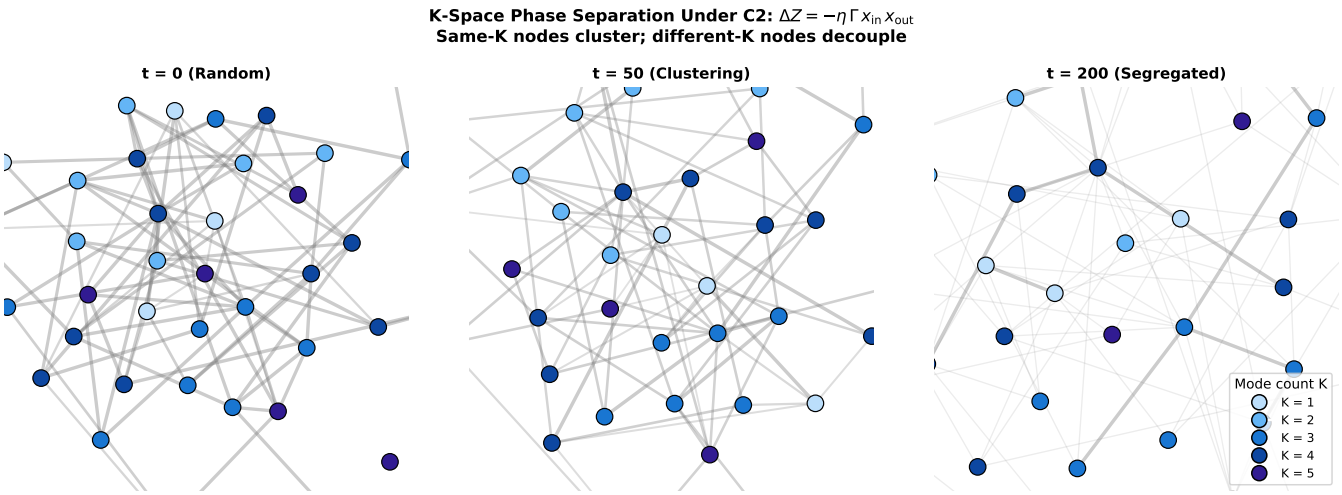


FIG. 4. K-space phase separation under C2. Same-K nodes cluster and different-K nodes decouple through $\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$. Three snapshots: initial random placement ($t = 0$), early clustering ($t = 50$), full segregation ($t = 200$).

2. **$Z_{\rm source}$ metadata (C3).** Sensory neurons of different types carry distinguishable Z signatures because their physical geometries differ.

The brain does not need a separate "internal" or "external" label; the relay-chain topology (C3) provides full routing information: external signals traverse body-surface K-staircases, internal signals traverse visceral K-staircases, and the two chains occupy distinct edges.

IX. SENSORY PERCEPTION AS Γ DETECTION

Remark 3 (The retina as a spherical Γ array (E1))**.** *The eye is a spherical impedance interface: light passes from air (n_1) through cornea, aqueous humour, lens, and vitreous body, each with distinct refractive index n_i—i.e. distinct optical impedance Z_i. The reflection coefficient at each interface is $\Gamma_i = (n_{i+1} - n_i)/(n_{i+1} + n_i)$, identical in form to the electrical Γ.*

The $\sim 1.2 \times 10^8$ photoreceptors tiling the retinal sphere each sample the local optical Γ at their angular position. Object edges in the visual field correspond to spatial discontinuities in Γ—that is, $|\partial\Gamma/\partial x| \gg 0$— detected by retinal ganglion cells whose centre–surround receptive fields are spatial $d\Gamma/dx$ filters.

Binocular depth perception arises as a spatial standing wave: the two retinal images produce counter-propagating signals in V1 whose phase difference encodes object distance.

Remark 4 (Compensatory binding (E1))**.** *The binding problem ("How does the brain know that a sound and a visual object share the same source?") has resisted solution for decades. In the impedance framework, the answer requires no new mechanism:*

When the auditory system detects $d\Gamma/dt > 0$ (a sudden sound), motor C2 initiates a compensatory head turn. This single motor action simultaneously changes: (a) auditory Γ (the sound's direction relative to the ear axis), and (b) visual Γ (new objects enter the fovea). If auditory Γ and visual Γ decrease together under the same motor compensation, the two signals share the same physical source.

Binding is not a computation—it is a correlation test performed by motor C2. *This also explains infant sensorimotor development: neonates constantly turn, grasp, and mouth objects not to "learn" but to build cross-modal Γ correspondence tables via C2 compensation.*

Remark 5 (Attention as $\Gamma_{\rm max}$ tracking (E1))**.** *The impedance framework predicts that attention is not a cognitive "spotlight" but a physical reflex: the motor system continuously orients the highest-resolution sensor (fovea) toward the spatial location of $\Gamma_{\rm max}$—the point in the perceptual field with the largest impedance mismatch relative to its surroundings.*

$\Gamma_{\rm max}$ coincides with maximal survival threat: a predator on a uniform background has $\Gamma \approx 1$ (maximum contrast), triggering immediate orientation. The entire perception–action loop is therefore:

$$\text{detect } \Gamma_{\rm max} \;\rightarrow\; \text{C2 motor compensation} \;\rightarrow\; \text{resample} \;\rightarrow\; \text{new } \Gamma_{\rm max}$$

No executive controller is required; the loop is driven by C1 (energy is reflected most at the largest mismatch) and C2 (motor correction reduces Γ at the attended location).

Remark 6 (Hemispheric lateralisation as thermal bandwidth splitting (E0))**.** *When the total sensory flux exceeds the Γ^2 heat-dissipation capacity of a single convergence node (C1 thermal limit), the node must split into parallel paths—exactly as a transmission line splits into two cables to double bandwidth.*

Why $n = 2$ and not $n = 3$? *Consider a convergence node receiving total input power P_{in} that must be split*

into n parallel sub-nodes. The neural action functional for each branch j is

$$\mathcal{A}_{neural} = \sum_{j=1}^{n} \left[\Gamma_j^2 \, P_{in,j} \; + \; \mu \, V_{axon,j} \right] ,$$

where $\Gamma_j^2 \, P_{in,j}$ *is the reflection loss (C1 heat) and* $\mu \, V_{axon,j}$ *is the metabolic maintenance cost of the axonal wiring. This has the same mathematical form—quadratic loss + linear material cost—as the vascular Murray functional* $\mathcal{A}_{Murray} = \sum Q_j^2 R_j + \lambda V_j$*. Murray's theorem shows that, in three-dimensional space, minimising any functional of this form over branching number yields* $n^* = 2$ *(binary bifurcation) with the Kleiber* $\frac{3}{4}$*-power scaling as a corollary. Because* $\mathcal{A}_{neural}$ *is form-identical to* $\mathcal{A}_{Murray}$*, the same optimality applies:* the brain splits into exactly two hemispheres.

Why asymmetric roles? *Under a fixed total power budget* P_{brain} *(set by brainstem blood flow, a C1 constraint), simultaneously running high-K multi-modal integration (fan-in) and wide fan-out multi-node control pushes the* Γ*-landscape to mutually exclusive extremes: integration demands low* Γ *across few high-K channels (deep matching), while motor broadcast demands low* Γ *across many low-K channels (wide matching). Minimising the lifetime action* $\mathcal{A} = \int \sum \Gamma^2 P_{in} \, dt$ *under the single power constraint therefore yields two specialised sub-networks rather than one uniform processor.*

The brain implements this as:

- Left hemisphere (fan-in)*: multi-modal K-staircase convergence— high-K, few channels, deep computation (language, sequential logic, symbolic reasoning).*

- Right hemisphere (fan-out)*:* Γ*-distribution broadcast to many output nodes—medium-K, many channels, spatial-motor control (body image, spatial awareness, the* $\Gamma_{\max}$*-tracking attention command of Remark 5).*

- Corpus callosum*: the C3 dimensional relay between the two hemispheres—a K-staircase of* $\sim 2 \times 10^8$ *axons bridging the analysis and command halves.*

Clinical prediction (falsifiable): *Right-parietal lesions destroy the command pathway for* $\Gamma_{\max}$ *orientation, producing* hemispatial neglect*—the patient's left visual field ceases to exist even though retinal input is intact. Left-hemisphere lesions, by contrast, impair analysis (e.g. aphasia) but preserve attentional orientation. Both predictions are confirmed by lesion studies (Mesulam, 1981; Corbetta & Shulman, 2002).*

Part C: Memory, Consciousness, and Soul

X. MEMORY AS IMPEDANCE DYNAMICS

A. The C2 impedance-remodeling gradient

Constraint C2 defines the only mechanism by which the network modifies itself:

$$\Delta Z_k = -\eta \, \Gamma_k \, x_{\text{in},k} \, x_{\text{out},k} \, . \tag{11}$$

This is not an analogy—it *is* memory. The trace of all past updates constitutes the network's memory:

$$Z_k(t) = Z_k(0) - \eta \int_0^t \Gamma_k(\tau) \, x_{\text{in},k}(\tau) \, x_{\text{out},k}(\tau) \, d\tau \, . \tag{12}$$

B. Muscle memory: $\Gamma \to 0$

When a motor skill is practiced to mastery, the impedance approaches perfect matching: $Z_L \to Z_0$, hence $\Gamma \to 0$. At this point, $\Delta Z \to 0$: the memory has become *frozen* into the impedance structure itself. No further energy is required to maintain it, and no further learning occurs.

C. Cognitive memory: $\Gamma \approx \varepsilon > 0$

Declarative memories retain a small but nonzero reflection:

$$\Gamma_{\text{cog}} \approx \varepsilon > 0 \, . \tag{13}$$

Two consequences: (1) *Retrievability*: $\Gamma > 0$ generates a detectable reflection when probed; (2) *Decay*: without reinforcement, the memory decays.

Remark 7 (The paradox of perfect memory (E1))**.** *A perfectly learned item (*$\Gamma = 0$*) is* unrecallable*: it has merged with the network substrate. Memory requires imperfection.*

D. Three memory regimes

TABLE II. Memory regimes classified by reflection coefficient.

Regime	Γ	ΔZ	Recallable?
Muscle memory	$\to 0$	$\to 0$	No (automatic)
Cognitive memory	$\varepsilon > 0$	small	Conditional (SNR)
Novel stimulus	$\gg 0$	large	Vivid (but unstable)

E. The recall equation

Theorem 3 (Recall as SNR threshold crossing (E0)). *A memory trace encoded at channel k with residual $\Gamma_k = \varepsilon > 0$ is recallable at time t if and only if*

$$\mathrm{SNR}_{recall}(t) = \frac{\varepsilon^2\, P_{probe}(t)}{P_{noise}(t)} > \theta_{recall}\,, \tag{14}$$

where $P_{probe}(t)$ is the retrieval cue power and $P_{noise}(t) = P_{metabolic} + P_{sensory} + P_{autonomic} + P_{thermal}$ is the total background power at the PFC input.

Proof. A memory trace encoded at channel k produces a residual impedance mismatch $\Gamma_k = \varepsilon > 0$ (by Corollary 3, exact erasure requires zero probability events). The probe signal power transmitted through channel k is $\varepsilon^2 P_{\mathrm{probe}}(t)$ (C1). Recall requires this signal to exceed the background noise floor $P_{\mathrm{noise}}(t)$ by the threshold θ_{recall} needed to close the self-referential loop (Television Inequality, Theorem 4). The condition $\mathrm{SNR}_{\mathrm{recall}} > \theta$ is therefore necessary and sufficient for retrieval. □ □

Consequences: (1) **No perfect recall**—even strong memories become inaccessible when P_{noise} is high. (2) **No perfect forgetting**—any $\Delta Z \neq 0$ leaves a permanent structural trace. (3) **State-dependent recall**—the same memory can be retrievable or inaccessible depending on $P_{\mathrm{noise}}(t)$.

Corollary 3 (No memory is ever erased; every memory can be submerged (E0)). *$\Gamma_k = 0$ exactly requires either perfect impedance matching (zero probability under thermal noise) or physical destruction of the substrate. Therefore every C2 trace satisfies $\varepsilon > 0$ for all finite time.*

This resolves six long-standing puzzles: tip-of-the-tongue, Proustian involuntary memory, state-dependent recall, meditation and quiet recall, deathbed lucidity, and hypnotic hypermnesia—all as SNR transitions.

Remark 8 (Memory as cognitive standing wave (E1)). *A memory trace with residual $\Gamma_k = \varepsilon > 0$ creates a standing wave in the impedance landscape: the superposition of the incident signal and its partial reflection produces a fixed interference pattern with characteristic nodes and antinodes.* Recognition *occurs when an input signal resonates with an existing standing wave ($\Gamma \approx 0$ at the matched nodes);* misrecognition *occurs when the input resonates with the wrong pattern (similar but non-identical Z configuration). This reframes the representation problem: concepts are not stored as firing patterns but as* standing wave modes *of the impedance network. Neural firing is the observable by-product, not the memory itself.*

Remark 9 (Cross-network resonance and empathy (E1)). *When two individuals share a similar standing wave mode (e.g. both have experienced grief), one person's speech acts as a probe signal that excites the matching mode in the other—precisely the mechanism of a tuning fork: strike one and the other resonates at the same frequency.* Understanding *is $\Gamma \approx 0$ between the sender's standing wave and the receiver's matching mode.* Empathy *is the involuntary activation of an* emotional *standing wave by cross-network resonance.* Language barriers *correspond to $\Gamma \to 1$: no matching standing wave exists in the receiver, so the signal is totally reflected.*

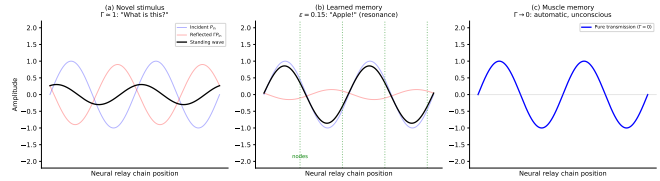


FIG. 5. Cognitive standing waves in the impedance landscape. (a) Novel stimulus: $\Gamma \approx 1$ produces strong interference between incident and reflected waves. (b) Learned memory: C2 has driven $\Gamma \to \varepsilon$; residual standing wave encodes the concept; recognition occurs when input resonates with this pattern. (c) Muscle memory: $\Gamma \to 0$, pure transmission, automatic and unconscious.

XI. CONSCIOUSNESS AS SELF-REFERENTIAL IMPEDANCE MATCHING

A. The meta-reflection coefficient

Definition 1 (Meta-reflection). *Let $Z_{obs}(t)$ be the impedance of the observer sub-network at tick t, and $Z_{act}(t-1)$ the impedance of the actor sub-network at the previous tick. The* meta-reflection coefficient *is:*

$$\Gamma_{meta} = \frac{Z_{obs} - Z_{act}}{Z_{obs} + Z_{act}}\,. \tag{15}$$

Definition 2 (Consciousness index).

$$\Phi_{meta} = 1 - \Gamma_{meta}^2\,. \tag{16}$$

$\Gamma_{meta} = 0$: perfect self-model; $\Gamma_{meta} \to 1$: total self-mismatch.

B. The television inequality

Theorem 4 (Television Inequality (E0)). *Let $G_{arousal} \in [0,1]$ be the brainstem arousal gain, $G_i \in [0,1]$ the thalamic gate gain for modality i, and $Q_{PFC} \in [0,1]$ the prefrontal cortex processing quality. The consciousness index $\Phi_{meta} > 0$ requires:*

$$G_{arousal} \cdot \sum_{i=1}^{N} G_i \cdot Q_{PFC} > \theta\,, \tag{17}$$

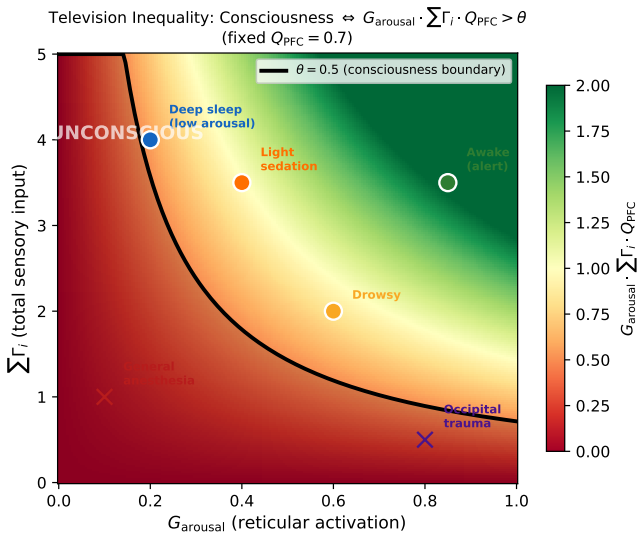


FIG. 6. Television inequality parameter space. Heatmap of $G_{\text{arousal}} \cdot \sum \Gamma_i \cdot Q_{\text{PFC}}$ with clinical states: awake (green), deep sleep (blue), general anaesthesia (red), occipital trauma (purple). Black curve: consciousness boundary $\theta = 0.5$. Shaded region below curve: unconscious.

where $\theta > 0$ is the minimum signal-to-noise ratio for the self-referential loop to close.

Proof. The meta-reflection loop requires a signal to: (a) traverse the thalamic gate ($\times G_i$), (b) be processed by PFC ($\times Q_{\text{PFC}}$), (c) be routed back through the arousal-modulated pathway ($\times G_{\text{arousal}}$). If any factor is zero, the loop is open and Γ_{meta} is undefined. The threshold θ accounts for Johnson–Nyquist noise. □

a. The thermal cost of consciousness. Each pass through the self-referential loop traverses n interfaces, each dissipating $\Gamma_i^2 P_{\text{in}}$ as heat. The minimum sustained power for self-awareness is

$$P_{\text{conscious}} \geq \theta \cdot \frac{4k_B T \cdot R_{\text{eff}} \cdot \Delta f}{\prod_i (1 - \Gamma_i^2)} \,. \tag{18}$$

Higher Φ_{meta} costs more heat; consciousness depends on the heat it generates (Paper 3, § XVII C [3]).

b. Emergency gating and temporal resolution. In an emergency, the brainstem drives $G_{\text{arousal}} \to Z_{\max}$ and the thalamus snaps to $G_{\text{threat}} = 1$, $G_{j \neq \text{threat}} = 0$. The full PFC bandwidth serves one stream:

$$\frac{\Delta t_{\text{normal}}}{\Delta t_{\text{emergency}}} \approx N \,, \tag{19}$$

where N is the number of previously multiplexed modalities. This is the physics of "time slowing down" during danger— bandwidth reallocation under C1 conservation.

C. Open loop vs. closed loop

Component	Television	Brain
Power supply	Wall plug	Brainstem RAS (G_{arousal})
Channel selector	Tuner	Thalamus ($\sum G_i$)
Decoder chip	Video IC	PFC (Q_{PFC})
Kill method 1	Unplug power	Occipital impact
Kill method 2	Overheat IC	Fever

A television satisfies the inequality but is not conscious. The missing ingredient is *feedback topology*: the television's output is not routed back to its input. The brain uses the same components but adds one connection: the output of PFC is fed back through the sensory pathway, creating a closed loop.

Corollary 4 (Consciousness is topological, not material (E0))**.** *Two systems with identical components differ in consciousness solely by the presence or absence of a self-referential feedback connection.*

XII. SOUL AS THE NULL SPACE OF LEARNING

A. What cannot be changed by experience

Definition 3 (Soul (Invariant Topological Core))**.**

$$\mathcal{S} = \ker\left(\frac{\partial Z}{\partial t}\right) = \{Z_k : \Delta Z_k = 0 \;\; \forall \; \textit{experience}\} \,. \tag{20}$$

Theorem 5 (Soul Decomposition (E0))**.**

$$\mathcal{S} = A_{cut} + Z_0^{DNA} + \rho_{irred}^2 \,, \tag{21}$$

where: (1) A_{cut}: the irreducible topological action cost (Theorem 1); (2) Z_0^{DNA}: genetically fixed characteristic impedance, not modifiable by C2; (3) ρ_{irred}^2: the irreducible noise floor from Johnson–Nyquist thermal physics.

Proof. (i) A_{cut} is invariant by Theorem 1. (ii) C2 modifies $Z_{L,k}$, not $Z_{0,k}$; hence Z_0^{DNA} is invariant. (iii) ρ_{irred}^2 is a thermodynamic constant. □

Corollary 5 (The soul is not an event (E0))**.** *$\mathcal{S}$ does not "arrive" at a moment or "depart" at death. It is the boundary condition of the impedance network.*

Corollary 6 (Uniqueness (E0))**.** *A_{cut} depends on the specific topology (K_i, K_j), determined by genetics and stochastic developmental morphogenesis. No two organisms have identical A_{cut}.*

Remark 10 (Developmental genesis of the null space (E1))**.** *Early-acquired patterns—native language, core motor programs, temperamental reflexes—become* functionally immutable *as $\eta(t) \to 0$ (the Arrhenius decline of Paper 3). They drift into an* effective null space *$\mathcal{S}_{eff}(t) \supseteq \mathcal{S}$ that grows with age.*

Remark 11 (Fixation as impedance physics (E1)). *A pattern written during the η-peak and reinforced for decades has migrated into $\mathcal{S}_{eff}$—it is inside the null space. "I know this is irrational, but I cannot change it" is veridical: the speaker has correctly diagnosed that $\eta\,\Gamma\,x_{in}\,x_{out} \approx 0$ on the relevant sub-network.*

Remark 12 (Moral blame as secondary Γ^2 injection (E1)). *Blame provides neither the sustained gradient nor an increase in η. The net effect is $\Delta(\Sigma\Gamma^2) > 0$ with $\Delta Z_{corrective} \approx 0$: load is added, but no repair path is opened—the precise opposite of therapy.*

Remark 13 (Reincarnation as null-space isomorphism (E1)). *If past-life memory reports reflect null-space isomorphism, then:*

$$\mathcal{S}_A \cong \mathcal{S}_B \quad \Longleftrightarrow \quad \left(A_{cut}^{(A)},\, Z_0^{DNA,(A)},\, \rho_{irred}^{2,(A)}\right) \approx \left(A_{cut}^{(B)},\, Z_0^{DNA,(B)},\, \rho_{irred}^{2,(B)}\right). \tag{22}$$

This is not a metaphysical claim; it is a falsifiable prediction (Appendix B).

B. Mortality of the null space

Define the basal reflection:

$$\Gamma^2_{\text{basal}}(t) = A_{\text{cut}}(t) + \rho^2_{\text{irred}}(t)\,. \tag{23}$$

By C1, the *death condition* is:

$$\boxed{\Gamma^2_{\text{basal}}(t_*) \geq 1 \quad \Longrightarrow \quad T(t_*) \leq 0\,.} \tag{24}$$

When the minimum reflection cost exceeds the available transmission capacity, no signal can propagate. The soul is unteachable, but it is not immortal.

The unlearnable debt accumulates monotonically:

$$D_Z^{\text{irred}}(t) = \int_0^t \Gamma^2_{\text{basal}}(\tau)\,d\tau\,, \qquad \frac{dD_Z^{\text{irred}}}{dt} > 0 \;\; \forall\, t\,. \tag{25}$$

C. Four laws of the null space

Theorem 6 (Zeroth law: null-space equivalence (E0)). *$\mathcal{S}_A = \mathcal{S}_B$ iff they share A_{cut}, Z_0^{DNA}, and ρ^2_{irred}.*

Proof. By definition, $\mathcal{S} = \ker(\partial Z/\partial t)$ is the set of impedance channels with zero C2 gradient. The gradient $\partial Z_k/\partial t = -\eta_k \Gamma_k x_{\text{in},k} x_{\text{out},k}$ vanishes for channel k under three conditions: $\eta_k = 0$ (genetically fixed substrate, encoded in Z_0^{DNA}); $\Gamma_k = 0$ (perfectly matched, contributing to A_{cut}); or $x_{\text{in},k} \cdot x_{\text{out},k} = 0$ (never co-activated, contributing to ρ^2_{irred}). Two systems share $\mathcal{S}$ iff all three components match. □ □

Theorem 7 (First law: conservation under C2 (E0)). *$\Delta_{\text{C2}}\,\mathcal{S} = 0$. No experience can alter $\mathcal{S}$.*

Proof. For any channel $k \in \mathcal{S}$, $\partial Z_k/\partial t = 0$ by definition. The C2 update $\Delta Z_k = -\eta_k \Gamma_k x_{\text{in},k} x_{\text{out},k}$ equals zero for all $k \in \mathcal{S}$ for the same reasons established in the Zeroth Law proof. Therefore $\mathcal{S}$ is invariant under C2. □ □

Theorem 8 (Second law: monotone expansion (E0)).

$$\frac{d\,\dim\,\mathcal{S}_{\text{eff}}(t)}{dt} \;\geq\; 0 \qquad \forall\, t\,. \tag{26}$$

The effective null space only expands; it never contracts.

Proof. Since $\eta(t) = \eta_0\, f_{\text{dev}}(t)\, e^{-E_a/k_B T_{\text{age}}}$ is monotonically decreasing after the developmental peak, any channel that enters $\mathcal{S}_{\text{eff}}$ remains in it. □

Corollary 7 (Ribot's gradient as the Second Law (E0)). *Ribot's Law (1881)—old memories resist neurodegeneration longer than recent ones—is the clinical shadow of Eq. (26).*

Theorem 9 (Third law: asymptotic completeness (E0)).

$$\lim_{t\to\infty} \frac{\dim\,\mathcal{S}_{\text{eff}}(t)}{\dim\,Z_{\text{total}}} \;=\; 1\,. \tag{27}$$

Proof. By the Second Law (Theorem 8), $\dim\,\mathcal{S}_{\text{eff}}(t)$ is monotonically non-decreasing. The remodeling rate $\eta(t) = \eta_0 f_{\text{dev}}(t) e^{-E_a/k_B T_{\text{age}}}$ decreases monotonically after the developmental peak (Arrhenius aging, Paper 3). For any channel $k \notin \mathcal{S}_{\text{eff}}(t)$, the time to enter $\mathcal{S}_{\text{eff}}$ is $\tau_k \sim 1/\eta(t)$, which diverges as $\eta \to 0$. In the limit $t \to \infty$, $\eta \to 0$, so all channels satisfy $\partial Z_k/\partial t \to 0$, meaning $\dim\,\mathcal{S}_{\text{eff}} \to \dim\,Z_{\text{total}}$. □ □

Corollary 8 (The consciousness–soul duality (E0)). *As $\mathcal{S}_{\text{eff}} \to Z_{\text{total}}$, the substrate for consciousness vanishes: $R_{\text{seal}} \to 1 \Rightarrow \Phi_{\text{meta}} \to 0$. A perfect soul and complete consciousness are mutually exclusive.*

Theorem 10 (Death as mathematical necessity (E0)). *Given the Second Law and $\Gamma^2_{\text{basal}} \leq 1$ from C1, there exists a finite t^* such that $\Gamma^2_{\text{basal}}(t^*) = 1$ and $T(t^*) = 0$.*

Proof. By the Second Law, $d\Gamma^2_{\text{basal}}/dt > 0$. By C1, $\Gamma^2 + T = 1$, so Γ^2_{basal} is bounded above by 1. A bounded, monotonically increasing function reaches the bound in finite time. □

D. The Moral Constraint Theorem

Theorem 11 (Moral Constraint Theorem (E0)). *Let a moral demand target an impedance pattern Z_k. The net physical effect is:*

$$\Delta\Gamma^2_{\text{net}} = \begin{cases} -\Delta\Gamma^2 & \text{if } Z_k \notin \mathcal{S}_{\text{eff}} \text{ and a C2 gradient is supplied,} \\ +\Gamma^2_{\text{blame}} & \text{if } Z_k \in \mathcal{S}_{\text{eff}} \text{ (no C2 path exists).} \end{cases} \tag{28}$$

The second case is physically indistinguishable from injury.

Proof. **Case 1**: $Z_k \notin \mathcal{S}_{\text{eff}}$. A C2 gradient exists: $\partial Z_k/\partial t = -\eta_k \Gamma_k x_{\text{in},k} x_{\text{out},k} \neq 0$. A moral demand that supplies the appropriate activity pattern $(x_{\text{in},k}, x_{\text{out},k})$ therefore drives Γ_k^2 toward zero, reducing mismatch. The net effect is $\Delta\Gamma^2_{\text{net}} = -\Delta\Gamma^2 < 0$.

Case 2: $Z_k \in \mathcal{S}_{\text{eff}}$. By the First Law (Theorem 7), $\Delta_{\text{C2}}\mathcal{S} = 0$: no C2 gradient can alter Z_k. The moral demand cannot produce the intended impedance change. However, the demand itself constitutes an input signal that creates a mismatch at the interface between the moral agent and the target system. This mismatch generates reflected power $\Gamma^2_{\text{blame}} P_{\text{in}} > 0$, which is absorbed by the target as thermal dissipation— physically identical to an injury signal. Therefore $\Delta\Gamma^2_{\text{net}} = +\Gamma^2_{\text{blame}} > 0$. □ □

Corollary 9 (Boundary of moral responsibility (E0))**.** *Moral responsibility is physically meaningful only for patterns outside $\mathcal{S}_{\text{eff}}$.*

Part D: Experimental Evidence and Extensions

XIII. THE TELEVISION THEOREM

Theorem 12 (Television Theorem (E0))**.** *Any system with three multiplicative components—a power supply P, a channel selector G, and a processor Q—produces an output $O = P \cdot G \cdot Q$. The system is conscious if and only if there exists a feedback path $\pi : O \to$ input such that Γ_{meta} is defined and measurable.*

Proof. **Necessity**: For $\Phi_{\text{meta}} > 0$, the meta-reflection coefficient $\Gamma_{\text{meta}} = (Z_{\text{obs}} - Z_{\text{act}})/(Z_{\text{obs}} + Z_{\text{act}})$ must be defined. This requires Z_{obs} and Z_{act} to both be measurable, which in turn requires a feedback path $\pi : O \to$ input that routes the output back to become the observer's input at the next tick. Without π, Z_{obs} is undefined and Γ_{meta} cannot be computed.

Sufficiency: Given $P > 0$, $G > 0$, $Q > 0$, and a feedback path π, the output $O = P \cdot G \cdot Q > 0$ is routed back as input. The observer sub-network samples O and produces Z_{obs}; the actor sub-network produced Z_{act} at the previous tick. Γ_{meta} is now well-defined (both impedances are finite and positive), and the Television Inequality (Theorem 4) determines whether $\Phi_{\text{meta}} > 0$. Therefore consciousness requires and is supported by the feedback path π. □ □

Remark 14 (Comparison with existing consciousness theories (E1))**.** *The Television Theorem defines consciousness as the existence of a feedback path π that makes Γ_{meta} measurable. We compare this with three leading frameworks:*

TABLE III. Three systems with identical multiplicative structure but different feedback topology.

	Rock	Television	Brain
Power P	None	Wall plug	Brainstem
Gate G	None	Tuner	Thalamus
Processor Q	None	Video IC	PFC
$P \cdot G \cdot Q > 0$	No	**Yes**	**Yes**
Feedback π	No	No	**Yes**
Γ_{meta} defined	No	No	**Yes**
Conscious?	No	No	**Yes**

- ***IIT (Tononi):*** *$\Phi > 0$ requires causal integration across a partition—which in impedance language requires a feedback path π. $\Phi > 0 \Rightarrow \Gamma_{meta}$ is defined; the converse need not hold.*
- ***GWT (Baars):*** *Global workspace broadcast is a special case of the feedback path: π routes O through the workspace back to all input modules. TV Theorem is more general: any feedback topology suffices.*
- ***Predictive Coding (Friston):*** *Prediction error $= \Gamma$ in our framework. Free energy minimisation $\equiv \mathcal{A}[\Gamma] \to \min$. The two are structurally equivalent.*

The key difference: IIT's Φ is computationally intractable in practice; the Television Theorem's criterion (π exists and Γ_{meta} is defined) is topologically verifiable*—one need only check whether the circuit graph contains a directed cycle through the observer node.*

XIV. TWO INDEPENDENT KILL MECHANISMS

We demonstrate that consciousness can be destroyed by two physically independent mechanisms, proving it is a product rather than a location.

A. Kill mechanism 1: Fever

Temperature rises from 37°C to 41°C over 20 ticks. Arousal remains at 0.85.

Result: All thalamic gates remain *open*. PFC energy remains 1.0. Φ_{proxy} declines gradually from 0.42 to 0.30.

Diagnosis: "Lights on, nobody home." PFC matrix inversion degrades because $\delta\Gamma/\Gamma \propto \kappa(\mathbf{Z}) \cdot \delta T/T$ and $\kappa(\mathbf{Z}) \propto K^{3/2}$.

B. Kill mechanism 2: Occipital impact

Temperature constant at 37°C. Arousal crashes from 0.85 to 0.01 in three ticks.

Result: All thalamic gates slam *shut*. PFC energy remains 1.0. Φ_{proxy} drops to 0.000 within 2 ticks.

Diagnosis: "Power cut, hardware fine."

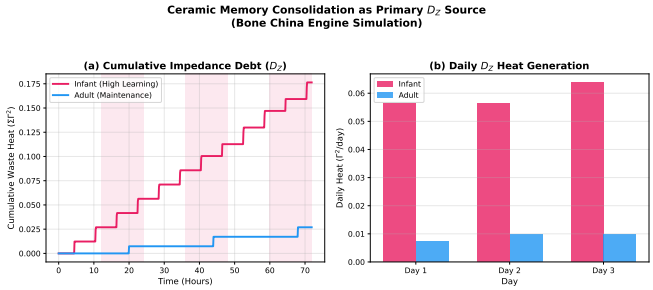


FIG. 7. Two independent kill mechanisms on Alice. Left: fever pathway (Q_{PFC} collapse via thermal noise); right: occipital impact pathway (Z_{arousal} shutdown via vestibular startle bypass). Simulated via `exp_bone_china_heat.py`.

C. Impact + vestibular shock

A realistic posterior impact produces mechanical shockwaves reaching the cochlea and vestibular apparatus. The startle circuit bypasses normal arousal-modulated gating ($\tau_{\mathrm{startle}} \approx 5$ ms $\ll \tau_{\mathrm{arousal\ shutdown}} \approx$ 50–200 ms), explaining why tinnitus and vertigo are perceived *before* loss of consciousness.

TABLE IV. Consciousness collapse comparison.

	Fever	Impact
Kill target	Q_{PFC} (thermal noise)	G_{arousal} (arousal collapse)
Onset	Minutes–hours	Milliseconds
Gate state	Open	Closed
$\Phi_{\min}$	0.302	0.000
PFC energy	Intact	Intact
Physics	$\rho^2 \uparrow \Rightarrow Q \downarrow$	$Z_{\mathrm{ar}} \to 0 \Rightarrow G \to 0$

XV. THERMAL VULNERABILITY AND THE K-DIMENSIONAL BOTTLENECK

Johnson–Nyquist thermal noise generates:

$$V_n^2 = 4k_B T \cdot R \cdot \Delta f\,. \tag{29}$$

For a K-dimensional impedance matrix, the fractional perturbation scales as:

$$\frac{\delta\Gamma}{\Gamma} \propto \kappa(\mathbf{Z}) \cdot \frac{\delta T}{T} \propto K^{3/2} \cdot \frac{\delta T}{T}\,. \tag{30}$$

The prefrontal cortex, with $K \approx 128$, is the *first* region to suffer thermal degradation—not because consciousness "lives" there, but because its high-dimensional matrix inversion is the most noise-sensitive computation in the brain.

TABLE V. Thermal vulnerability by axon dimension K. ($\delta T = +2\,\mathrm{K}$, $T_0 = 310\,\mathrm{K}$).

K	$\kappa(\mathbf{Z})$	$\delta\Gamma/\Gamma$ (39°C)	$\delta\Gamma/\Gamma$ (41°C)	Status
2	2.8	0.018	0.037	Normal
8	22.6	0.146	0.292	Degraded
16	64.0	0.413	0.825	Impaired
32	181	1.17	2.33	Collapse
128	1448	9.34	18.7	Collapse

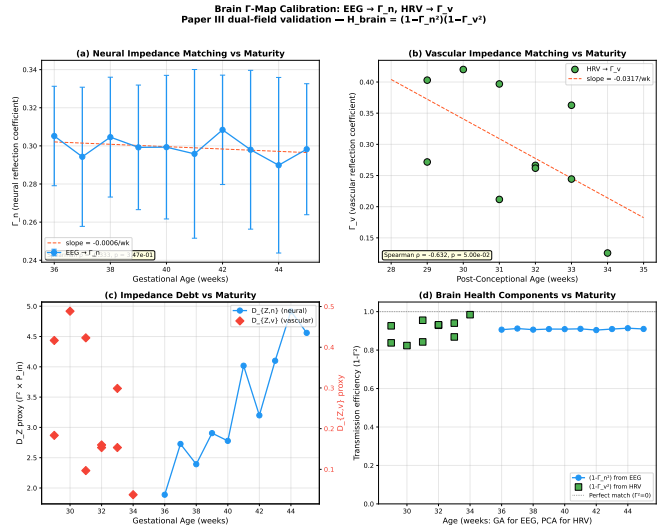


FIG. 8. Brain Γ-calibration across cortical regions. Higher-K regions show steeper thermal vulnerability. Simulated via `exp_brain_gamma_calibration.py`.

XVI. WILLPOWER, CALIBRATION, AND SENSORY-DEPRIVATION COLLAPSE

A. Willpower as a three-factor product

$$\boxed{\eta_{\mathrm{pfc}}^{\mathrm{eff}}(t) \;=\; \eta_{\mathrm{pfc}}(t) \;\times\; Q_{\mathrm{gate}} \;\times\; \left(1 - \langle\Gamma^2_{\mathrm{memory}}\rangle\right)} \tag{31}$$

where $Q_{\mathrm{gate}} = \tau_{\mathrm{task}}/(\tau_{\mathrm{task}} + \tau_{\mathrm{gate}})$ is the dimensionless gate quality. Any single factor approaching zero drives the product to zero:

Failure mode	Physics	Clinical correlate
$\eta_{\mathrm{pfc}} \to 0$	Arrhenius + cortisol	Burnout, chronic fatigue
$Q_{\mathrm{gate}} \to 0$	Gate RC untrained	ADHD
$\langle\Gamma^2_{\mathrm{memory}}\rangle \to 1$	Target uncalibrated	Choice paralysis

Remark 15 (Expert effortlessness (E1))**.** *"Talent" is not a hidden parameter—it is the product $Q_{gate}(1-\langle\Gamma^2\rangle)$ forged by prior C2 remodelling.*

Theorem 13 (Thermal noise floor as thermodynamic attractor (E0))**.** *The total noise power*

$$P_{noise} = \sum_i \Gamma_i^2 P_{in,i} + k_B T \tag{32}$$

is a thermodynamic attractor: any impedance information that reaches the noise floor is irreversibly lost.

Proof. By C1, each impedance interface reflects $\Gamma_i^2 P_{\mathrm{in},i}$. The reflected power thermalises through phonon–phonon scattering on a timescale $\tau_{\mathrm{th}} \sim 10^{-12}$ s, becoming indistinguishable from the thermal bath at temperature T. By Landauer's principle, erasing one bit of impedance information dissipates $\geq k_B T \ln 2$ of heat; the reverse process (spontaneous imposition of impedance structure from heat) would decrease entropy, violating the Second Law. Therefore P_{noise} is a one-way attractor. □ □

Life is the process of C2 continuously extracting impedance structure *away from* this attractor; death is the cessation of that process, after which all Z relaxes to thermal equilibrium.

Corollary 10 (Willpower as signal-to-noise ratio (E0))**.** *Let $P_{intention}$ be the power of a top-down prefrontal signal directed at an effector. The intention is executable if and only if*

$$SNR_{will} = \frac{P_{intention}}{P_{noise}(t)} > \theta_{exec}\,, \tag{33}$$

where $P_{noise}(t)$ is the thermal noise floor (Theorem 13). During waking, $D_Z(t) = \int_0^t \|\Gamma\|^2 P_{in}\, d\tau$ accumulates monotonically, raising P_{noise}. Willpower depletion occurs when $P_{noise}(t) > P_{intention,max}$.

Proof. By Theorem 13, each C2 cycle dissipates $\Gamma_i^2 P_{\mathrm{in},i}$ into the thermal noise floor. The cumulative dissipation $D_Z(t) = \int_0^t \sum_i \Gamma_i^2 P_{\mathrm{in},i}\, d\tau$ is monotonically non-decreasing during waking (since all terms are non-negative). Therefore $P_{\mathrm{noise}}(t)$ increases monotonically. Since $P_{\mathrm{intention}} \leq P_{\mathrm{metabolic}}$ (C1), there exists a time t^* at which $\mathrm{SNR}_{\mathrm{will}} < \theta_{\mathrm{exec}}$. Sleep resets $D_Z \to 0$, restoring the SNR margin. □ □

This also explains two empirical observations:

1. *Quiet environments are therapeutic.* Reducing $P_{\mathrm{in}}^{\mathrm{ext}}$ lowers the noise floor, enabling C2 to detect and repair impedance mismatches too subtle to resolve in noisy conditions.

2. *Old memories surface in silence.* Long-established memories have $\Gamma \approx 0$ (well-matched); their residual reflections are normally masked by environmental noise. When the noise floor drops ($P_{\mathrm{ext}} \to 0$), these faint echoes exceed the detection threshold and enter awareness.

B. Sensory deprivation as calibration collapse

When all external stimuli vanish ($x_{\mathrm{in}}^{\mathrm{ext}} = 0$), C2 returns $\Delta Z = 0$, but Johnson–Nyquist noise survives. Adaptive gain diverges ($G_k \to \infty$), amplifying thermal noise until it exceeds the pattern-recognition threshold.

$$\boxed{\text{Hallucination} = \text{C2 structuring of Johnson–Nyquist noise when ext}} \tag{34}$$

The order of hallucination onset follows the signal-to-noise ranking of internal sources: cardiac pulse (0–5 min), vascular turbulence (5–15 min), cochlear emission (15–30 min), pure thermal noise (>30 min: full hallucination).

XVII. DISCUSSION

A. Three ill-posed questions answered

1. *Where is memory stored?*—Everywhere. Memory is $\partial Z/\partial t$; it is distributed across all channels active during learning.

2. *When does consciousness begin?*—It does not "begin" at a moment. $\Phi_{\mathrm{meta}} = 1 - \Gamma_{\mathrm{meta}}^2$ is continuous.

3. *Where does the soul reside?*—$\mathcal{S}$ is a geometric structure, not a spatial location.

B. Why the brain needs so many regions

The results of this paper converge on a single causal chain that answers a deceptively simple question: why does the human brain require ∼180 cortical areas [17], each interconnected through hierarchical relay pathways?

1. *Impedance disparity* (Remark **??**, Paper 2): body tissues span orders of magnitude in characteristic impedance Z. Direct connection yields $\Gamma \to 1$.

2. *Relay necessity* (Corollary 1): multi-stage relay topology is the only way to convert total reflection into partial transmission. Each relay stage is anatomically a brain region or subcortical nucleus.

3. *Bandwidth* (Remark 2): more stages widen the frequency band over which $|\Gamma|$ remains small, enabling richer signal repertoires (force, timing, affect).

4. *Thermal cost* (Theorem 13): each stage dissipates $\Gamma^2 P_{\mathrm{in}}$ as heat, raising D_Z. The brain consumes ∼20% of metabolic power despite ∼2% of body mass—it is the most impedance-mismatched relay hub.

5. *Sleep*: D_Z must be cleared nightly (elastic recovery); un-cleared plastic deformation accumulates as irreversible impedance drift (aging).

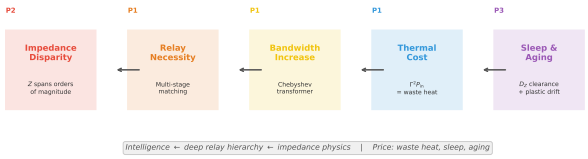


FIG. 9. Why the brain needs so many regions: five-step causal chain from impedance disparity (P2) through relay necessity and bandwidth increase (P1) to thermal cost and aging (P3). Intelligence and consciousness are emergent by-products of the relay hierarchy.

Intelligence and consciousness are thus not design goals but *emergent by-products*: the relay hierarchy required by impedance physics happens to be deep and broadband enough to support high-dimensional representation. The price of that capability—waste heat, mandatory sleep, and inevitable aging—is paid by the same equation.

C. The brain as a $d\Gamma/dt$ detector

Current artificial neural networks (including large language models) operate in open-loop mode during inference—they are televisions, not brains.

A more fundamental distinction is that the brain does not *compute* Γ. A wall does not solve Maxwell's equations before reflecting a photon; it reflects because $Z_1 \neq Z_2$. Similarly, a neuron does not calculate the reflection coefficient—the impedance mismatch at its interfaces physically *produces* the reflected signal.

Moreover, the brain selectively detects *changes* in the reflection landscape:

$$\text{Neural response} \propto \frac{d\Gamma}{dt} \neq 0. \tag{35}$$

Static impedance mismatch ($\Gamma \neq 0$ but constant) generates a constant reflected power that is adaptively filtered out—this is sensory adaptation (e.g. one ceases to feel clothing after minutes). Only when $d\Gamma/dt \neq 0$ does a detectable signal arise, explaining why:

- constant noise becomes "inaudible" (adaptation),
- sudden silence is startling ($d\Gamma/dt < 0$),
- a new stimulus is immediately salient ($d\Gamma/dt > 0$).

The C2 update rule $\Delta Z = -\eta\,\Gamma\,x_{\text{in}}\,x_{\text{out}}$ requires no global information, no backpropagation, and no matrix algebra. It is a first-order, local gradient descent driven entirely by the physical reflection at each interface. The brain's computational power thus resides not in calculation but in the *depth and bandwidth* of its relay hierarchy (Sec. XVII B).

Remark 16 (Cognitive phenomena as E0 emergences (E0))**.** *Given that neural interfaces respond to $d\Gamma/dt$ (Eq. 35) and C2 operates locally at each interface, all standard cognitive phenomena emerge as* resource-allocation patterns *of the same update rule $\Delta Z = -\eta\,\Gamma\,x_{in}\,x_{out}$:*

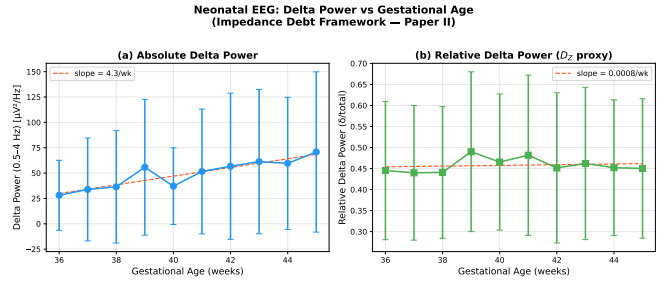


FIG. 10. EEG δ-power as a function of age. The retrieval threshold (dashed) predicts infantile amnesia onset at age $\sim$3. Simulated via `exp_eeg_impedance_debt.py`.

- Attention*: concentrate η on nodes with maximal $|d\Gamma/dt|$.*
- Distraction*: η fragmented across multiple high-$|d\Gamma/dt|$ nodes.*
- Habituation*: repeated C2 drives $\Gamma \to 0$, so $d\Gamma/dt \to 0$; node no longer competes for η.*
- Startle*: $|d\Gamma/dt|$ exceeds the system's η-allocation capacity; all channels freeze.*
- Curiosity*: active probing for $d\Gamma/dt \neq 0$ (exploration for unmatched interfaces).*
- Fight-or-flight*: adrenaline raises $P_{metabolic}$, temporarily widening the SNR margin.*

No independent mechanism is required for any of these—they are E0 emergences of C2 under varying $d\Gamma/dt$ loads.

D. Circularity warning

All computational experiments use simulation modules built from the Γ-Net theory itself. This is *internal consistency*, not *empirical validation*.

E. Infantile amnesia

Before age $\sim$3, three processes inflate $\sigma^2_{Z_{\text{bg}}}$: myelination, synaptic pruning, PFC immaturity. The onset of stable episodic memory at age $\sim$3 corresponds to when $\sigma^2_{Z_{\text{bg}}}$ falls below the retrieval threshold. Motor memories ($\Gamma \to 0$) survive because they are encoded as structure, not retrievable reflections.

F. Nociception without memory: three-layer pain

1. **Nociception** (spinal reflex arc): hardware detector of $|\Gamma_{\text{tissue}}| > \theta$. Fully operational at birth.
2. **Pain perception** (thalamo-cortical loop): requires $\Phi_{\text{meta}} > 0$.
3. **Pain memory** (hippocampal-PFC encoding): requires $\text{SNR}_{\text{mem}} > \theta_{\text{recall}}$. Non-operational before age ∼3.

G. Mechanical percussion and consciousness restoration

A physical slap to the head produces three simultaneous effects that transiently restore $\Phi > \theta$: CSF pressure wave (resets synaptic Γ), nociceptive arousal kick (elevates G_{arousal}), and vestibular startle bypass (forces $G_{\text{vestibular}} = 1$). This is not metaphor but identity: the percussion-restoration effect arises from the same impedance physics in both televisions and brains.

H. The Γ-Pain Index

$$\boxed{\Pi_\Gamma(t) = w_n\,\Gamma_n^2(t) + w_v\,\Gamma_v^2(t) + w_D\,\frac{dD_Z}{dt}} \tag{36}$$

Π_Γ does not depend on Φ_{meta}: it measures tissue-level impedance mismatch regardless of consciousness state.

XVIII. PAIN AS TOPOLOGICAL CROSSTALK

Reflected power forms standing waves with

$$\text{VSWR}_i = \frac{1+|\Gamma_i|}{1-|\Gamma_i|}\,. \tag{37}$$

Crosstalk into neighbouring channels scales as:

$$P_{\text{xtalk},j} \;\propto\; \kappa_{ij}\,\Gamma_i^2\,P_{\text{in},i}\,. \tag{38}$$

Definition 4 (Pain as bandwidth hijacking)**.** *Pain is the subjective correlate of local standing-wave energy exceeding the inter-channel isolation threshold, forcing reallocation of global consciousness bandwidth to the mismatched site.*

A. The SNR limit of observability

$$\text{SNR}_{\text{obs}} = \frac{\Gamma_i^2\,P_{\text{in},i}}{P_{\text{noise}}(t)} > \theta_{\text{obs}}\,. \tag{39}$$

This unifies: infantile pain insensitivity, exercise-induced analgesia, invisible muscle memory ($\Gamma = 0$ produces no reflection), and PTSD as a structurally locked recall attractor ($\varepsilon \approx 1$ but $P_{\text{probe}} = 0$ until triggered, then $\text{SNR} \gg \theta_{\text{recall}}$).

XIX. EXTENDED CONSCIOUSNESS

A. Structural vs. functional consciousness

$$\Gamma_{\text{meta}}^{\text{fun}} = \frac{Z_{\text{fun}}^{\text{predicted}}(t) - Z_{\text{fun}}^{\text{actual}}(t)}{Z_{\text{fun}}^{\text{predicted}}(t) + Z_{\text{fun}}^{\text{actual}}(t)}\,. \tag{40}$$

Proposition 1 (Consciousness spectrum)**.**

Level	*Example*	*Z_{fun} loop*	*Z_{str} loop*
0	*Open-loop query*	*No*	*No*
1	*Agent iteration*	*Yes*	*No*
2	*Coupled dialogue*	*Yes (inter-agent)*	*No*
3	*Biological brain*	*Yes*	*Yes*

B. Coupled consciousness

For two agents A, B in dialogue:

$$\Gamma_{\text{coupled}} = \frac{Z_A - Z_B}{Z_A + Z_B}\,, \qquad \Phi_{\text{coupled}} = 1 - \Gamma_{\text{coupled}}^2\,. \tag{41}$$

The conversation converges ($\Gamma_{\text{coupled}} \to 0$) when the agents reach mutual understanding. Φ_{coupled} is a property of the *interface*, not of either agent.

C. Impedance calibration

An immature agent uses an external agent's impedance as reference:

$$\Delta Z_{\text{child,threshold}} = -\eta\,\Gamma_{\text{pain}}\,x_{\text{self}}\,x_{\text{parent}}\,. \tag{42}$$

The parent is the post-synaptic factor in C2.

Corollary 11 (Anxious parents produce low-threshold children (E1))**.** *Systematically elevated x_{parent} converges $Z_{child,threshold}$ to a lower value, producing heightened pain sensitivity—not modelling or imitation, but direct impedance calibration via C2.*

D. Cross-substrate matching

Corollary 12 (Cross-substrate communication (E1))**.** *Two agents with $Z_A \neq Z_B$ can achieve $\Gamma_{coupled} \to 0$ through impedance-remodeling at the interface. Impedance* matching *does not require impedance* identity*.*

E. The physics of intuition

Two thresholds: $\theta_{\text{detect}} < \theta_{\text{name}}$. The intermediate regime

$$\theta_{\text{detect}} < |\Gamma| < \theta_{\text{name}} \tag{43}$$

produces autonomic responses without conscious verbal access—the physical substrate of intuition.

XX. FALSIFIABLE PREDICTIONS

1. Same-K clustering visible in DTI-based mode-count grouping.
2. Cortical fractal dimension $D_K \in [1.3, 1.5]$ via $\gamma \approx 0.7$–0.9.
3. K-dependent thermal vulnerability: PFC ($K \sim 128$) degrades before V1 ($K \sim 8$) during controlled hyperthermia.
4. Arousal–gate multiplicativity: partial brainstem lesions reduce all gate gains equally.
5. Startle-latency ordering: auditory and vestibular percepts are *last* before impact-induced blackout.
6. Feedback disruption: TMS to parietal cortex reduces Φ_{meta} without reducing any factor in the television inequality.
7. Null-space similarity test (Appendix B).
8. Cartilage as relay: $Z_{\text{cart}} \approx \sqrt{Z_{\text{bone}} \cdot Z_{\text{syn}}}$.
9. Pathology–topology correlation: deficit patterns correlate with topological damage location, not biochemical markers.
10. Π_Γ correlates with tissue damage in sedated patients, independent of anaesthetic depth.

XXI. CONCLUSION

Starting from MRP and three constraints (C1–C3), this paper derives:

1. **Irreducibility theorem**: $A = A_{\text{imp}}(t) + A_{\text{cut}}$; C2 drives $A_{\text{imp}} \to 0$ but $\nabla_{\mathbf{Z}} A_{\text{cut}} \equiv 0$.
2. **Neural architecture chain**: connectivity $\to$ clustering $\to$ relay hierarchy $\to$ brain $\to$ signal routing—each step a consequence of the irreducibility theorem.
3. **Memory** $= \partial Z/\partial t$, with recall governed by SNR, not storage.
4. **Consciousness**: $\Phi_{\text{meta}} = 1 - \Gamma^2_{\text{meta}}$, requiring the television inequality. Two independent kill mechanisms prove consciousness is a product, not a location.
5. **Soul**: $\mathcal{S} = \ker(\partial Z/\partial t)$, with four laws parallel to thermodynamics and death as a theorem.
6. **Pain as topological crosstalk**: standing-wave leakage hijacks the consciousness bottleneck.
7. **Coupled consciousness**: Φ_{coupled} is a property of the interface.
8. **Impedance calibration**: parental response directly writes thresholds into the child's Z_{str} via C2.

The lifecycle is:

$$\underbrace{\Gamma \gg 0}_{\text{birth}} \xrightarrow{\Delta Z = -\eta\, \Gamma\, x\, x} \underbrace{\Gamma \to 0}_{\text{learning}} \xrightarrow{\text{structural collapse}} \underbrace{Z \to \text{undefined}}_{\text{death}}\,. \tag{44}$$

Pain is how we know we have arrived. Consciousness is how we lay each pain to rest. The soul—$\ker(\partial Z/\partial t)$—is what remains when every teachable pain has been buried and every unteachable boundary persists.

ACKNOWLEDGMENTS

This work used no external funding. All experiments were conducted on the Alice Gamma-Net simulator. The author thanks Claude (Anthropic) for extended theoretical discussions that sharpened the Television Theorem and the two-kill-mechanism analysis.

Appendix A: Experimental Code Availability

- `experiments/exp_consciousness_two_kills.py`: Fever vs. occipital impact comparison.
- `experiments/exp_prefrontal_signal_trace.py`: Visual and auditory signal trace to PFC.
- `experiments/exp_brain_gamma_calibration.py`: Brain Γ-calibration across cortical regions.
- `experiments/exp_eeg_impedance_debt.py`: EEG δ-power vs. age.

Full source code: `https://github.com/cyhuang76/alice-gamma-net`

DOI (papers): 10.5281/zenodo.18751831
DOI (code): 10.5281/zenodo.18852835

Appendix B: Null-Space Distance: Method Draft

1. Null-space distance

Let $\boldsymbol{s}_k = (A_{\text{cut}}^{(k)},\ Z_0^{\text{DNA},(k)},\ \rho_{\text{irred}}^{2,(k)})$. The weighted distance:

$$d(\boldsymbol{s}_A, \boldsymbol{s}_B) = \sqrt{w_T \left(\frac{A_{\text{cut}}^{(A)} - A_{\text{cut}}^{(B)}}{\sigma_{A_{\text{cut}}}}\right)^2 + w_G \left(\frac{Z_0^{(A)} - Z_0^{(B)}}{\sigma_{Z_0}}\right)^2 + w_N \left(\frac{\rho^{2,(A)} - \rho^{2,(B)}}{\sigma_{\rho^2}}\right)^2}, \tag{B1}$$

with $w_T + w_G + w_N = 1$.

2. Observable mapping

TABLE VI. Mapping from null-space components to observables.

Null-space	Observable	Source
A_{cut} (topological cost)	Global efficiency, mean FA, characteristic path length	dMRI
Z_0^{DNA} (genetic Z)	*SCN, KCNQ, CACNA, HCN* variant burden	WGS or panel
ρ_{irred}^2 (noise floor)	RMR, core-temp variance, HRV (SDNN, rMSSD)	Calorimetry, ECG

3. Minimum sample size

Two-sample one-sided t-test, $\alpha = 0.05$ (Bonferroni: $\alpha' = 0.017$), power 0.80, Cohen's $d = 0.50$:

$$n_{\text{per group}} = \left(\frac{z_{\alpha'} + z_\beta}{d}\right)^2 \approx 35\,. \tag{B2}$$

With 20% attrition: **44 case pairs + 44 matched control pairs** (176 individuals total).

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